

ARTICLE

Open Access

A nonlinear stiffness softening mechanism for low-bias, high-sensitivity MEMS accelerometers with extended dynamic range

Peyman Firoozy^{1✉}, Milad Seifnejad Haghighi¹, Mikhail Kanygin¹, Philip Ferguson² and Behraad Bahreyni^{1✉}

Abstract

The small dimensions of microfabricated devices present challenges in applications such as inertial sensing, where a larger proofmass is necessary for enhanced sensitivity. An effective approach to addressing the limitations of linear sensing is to use nonlinear mechanisms that adapt the device's response according to different operating conditions. This paper introduces a new nonlinear spring mechanism for use in microsensors that harnesses the buckling phenomenon to achieve stiffness softening. The proposed mechanism utilizes a micro-arm to apply an eccentric axial load to an inclined beam, causing it to buckle in a controlled manner under a specified load. Once buckled, linear springs dominate the response of the system. We demonstrate that this method results in a smaller bias displacement compared to previously reported techniques based on snap-through behaviour, leading to potential reductions in device size and improvements in operational range. The behaviour is analytically modelled and verified through simulations. A prototype device was designed and microfabricated to experimentally validate the design principles. Compared to pre-curved nonlinear springs, the proposed design results in an 11-fold reduction in bias force, a 100-fold reduction in bias displacement, and a reduction in mechanical stiffness by a factor of 520. These results were verified through experiments conducted on a microfabricated accelerometer with an on-chip optical interferometer. Test results reveal an extended linear range of better than 150 mg, a bias force of 0.3 mN, and a bias displacement of 10 μm , measured with an integrated optical interferometer with a displacement noise floor of 40 pm/ $\sqrt{\text{Hz}}$ at 2 Hz and sensitivity of 194°/mg.

Introduction

Nonlinear spring mechanisms, including geometric anti-springs (GAS) and bistable pre-curved beams, have garnered significant interest in precision sensing applications due to their capability to reduce the effective spring constant of suspension structures. These mechanisms are particularly valuable in developing high-resolution Micro-Electro-Mechanical Systems (MEMS) accelerometers for applications such as navigation¹, structural health monitoring², gravity measurement³,

seismic sensing⁴, and space exploration⁵, which require low noise, low power consumption, compactness, and cost-effectiveness for wide-scale deployment. In these applications, sensitivity, noise, and measurement range are the key performance parameters of the sensor. Several strategies have been explored to enhance sensitivity, such as increasing proofmass size^{6–8}, employing optical sensing schemes^{9–11}, and reducing stiffness^{10,12–15}. Notably, Middlemiss et al.¹⁶ utilized snap-through behavior in three curved beams to create an anti-spring effect, leading to the development of a MEMS accelerometer capable of detecting Earth tides, achieving a noise floor of 40 ng/ $\sqrt{\text{Hz}}$ at 1 Hz ($1\text{g} \approx 9.81\text{ ms}^{-2}$) with a bias displacement of 800 μm . The same research group¹⁷ later developed a symmetrical four-anti-spring mechanism, with a resonant frequency of 7.35 Hz and a noise floor of

Correspondence: Peyman Firoozy (pfa24@sfu.ca) or Behraad Bahreyni (bba19@sfu.ca)

¹School of Mechatronic Systems Engineering, Simon Fraser University, Surrey, BC V3T 0A3, Canada

²Department of Mechanical Engineering, University of Manitoba, Winnipeg, MB R3T 5V6, Canada

© The Author(s) 2025



Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

8 ng/ $\sqrt{\text{Hz}}$, making it suitable for field-portable MEMS accelerometers. Boom et al.¹⁸ designed an ultra-sensitive MEMS accelerometer using anti-spring structures, achieving a noise floor below 2 ng/ $\sqrt{\text{Hz}}$ at 28 Hz. El Mansouri et al.¹⁹ enhanced their MEMS accelerometer sensitivity by incorporating a mechanical anti-spring effect and an application-specific integrated circuit (ASIC), resulting in a theoretical noise floor of 17 ng/ $\sqrt{\text{Hz}}$. Their design reports a minimum resonant frequency of 8.7 Hz with a required bias displacement of 600 μm . Zhang et al.²⁰ implemented anti-springs in a MEMS accelerometer, achieving a noise floor of 52 ng/ $\sqrt{\text{Hz}}$ at 1 Hz and a bandwidth of 158 Hz. Additionally, Tang et al.^{21,22} used a combination of curved beams as negative springs and folded beams as positive springs to generate a quasi-zero stiffness effect, which resulted in a noise floor of 8 ng/ $\sqrt{\text{Hz}}$ and a large bias displacement of 500 μm . Weidong et al.²³ utilized varying pre-curved beam widths to achieve a low resonant frequency across different tilt angles, reaching a noise floor of 9.2 ng/ $\sqrt{\text{Hz}}$ at 6 Hz and bias displacement of 350 μm . Xiong et al.²⁴ designed two parallel pre-curved connected beams as an anti-spring for sensitivity enhancement in MEMS accelerometers. Their measurement results show a noise floor of 21.3 $\mu\text{g}/\sqrt{\text{Hz}}$ with a small bias displacement of 1.6 μm . However, this small bias displacement comes at the cost of a high resonant frequency of 2.3 kHz. Wu et al.²⁵ introduced a free-form anti-spring mechanism, reporting a low resonant frequency of 1.71 Hz and a noise floor of 1.1 ng/ $\sqrt{\text{Hz}}$ at 1 Hz. Achieving this low resonant frequency requires a bias displacement of 1000 μm . However, these approaches require precise matching of the positive and negative spring constants, where even minor discrepancies can prevent the system from achieving near-zero stiffness.

Achieving high sensitivity for MEMS accelerometers is often at the cost of dynamic range due to the inherent tradeoff between sensitivity and working range. Consequently, the most sensitive devices exhibit dynamic ranges as low as few mg, severely limiting their application. To enhance the dynamic range, many researchers have adopted closed-loop systems in MEMS accelerometers, which apply a feedback force to counteract the inertial forces acting on the proofmass using electrostatic²⁶ or electromagnetic actuation²⁷. However, MEMS accelerometers with high sensitivity typically use large proofmasses. As a result, high acceleration inputs require a substantial feedback force to balance the inertial force, necessitating either a high actuation voltage or a large array of actuation capacitors²⁸. Nevertheless, this approach presents significant design, fabrication, and control challenges, especially when integrating more proofmasses into a single module. In comparison, optical MEMS accelerometers have gained

significant attention due to their high displacement accuracy and inherent robustness against electromagnetic interference, making them particularly suitable for sensing in harsh environments²⁹.

To address the trade-off between sensitivity and dynamic range, this work introduces a new nonlinear spring mechanism based on an inclined beam that buckles under a preset axial load. Theoretical modeling and finite element analysis demonstrate that adjusting the inclined beam angle and the length of the connected micro-arm can significantly reduce the bias force and displacement to achieve stiffness softening regime. A prototype device, in the form of a spring accelerometer, was designed, microfabricated, and characterized using an on-chip optical interferometer to validate the design principles and models. The proposed structure can be incorporated into high-performance micromachined sensors that need to operate in the presence of a strong DC interference, such as gravimeters. Its small bias displacement enables the use of alternative device interfaces, such as capacitive sensing, for operation.

Results

Modeling and design of the nonlinear-spring mechanism

Consider an ideal elastic Euler beam subjected to a centrally compressive axial load. Initially, the beam is stiff and responds linearly without any transverse deflection. However, once the applied force exceeds a critical threshold, determined by the beam's dimensions and boundary conditions, the beam becomes unstable in its original configuration and adopts one of two symmetric bent shapes, a phenomenon known as pitchfork bifurcation, as shown in Fig. 1a, b. Buckled beam structures are attractive in microsystem designs due to their simplicity and ability to generate large displacements with minimal restoring force. This behavior has been exploited in a wide range of applications, including energy harvesters³⁰, morphable microelectronic structures³¹, switches³², resonators³³, sensors³⁴, and actuators³⁵.

In such applications, particularly in the design presented here, controlling the direction of buckling is essential. To address this challenge, we implemented eccentric loading using a micro-arm with a length of d connected to inclined beams with an angle of θ (Fig. 1c). This configuration removes one of the two equilibrium states, ensuring that the beam always buckles in a pre-determined direction. Since buckling is a rapid unstable transition, linear spring with stiffness k is connected to the beam to control the transition speed. A vertical inertial force in y-direction, F , and moment, M , due to the micro-arm, are applied to the tip of the inclined beam, which is assumed to have a uniform cross-section with dimensions of thickness t , width w , and length L_i . Figure 1d illustrates the free body diagram (FBD) of the mechanism in the

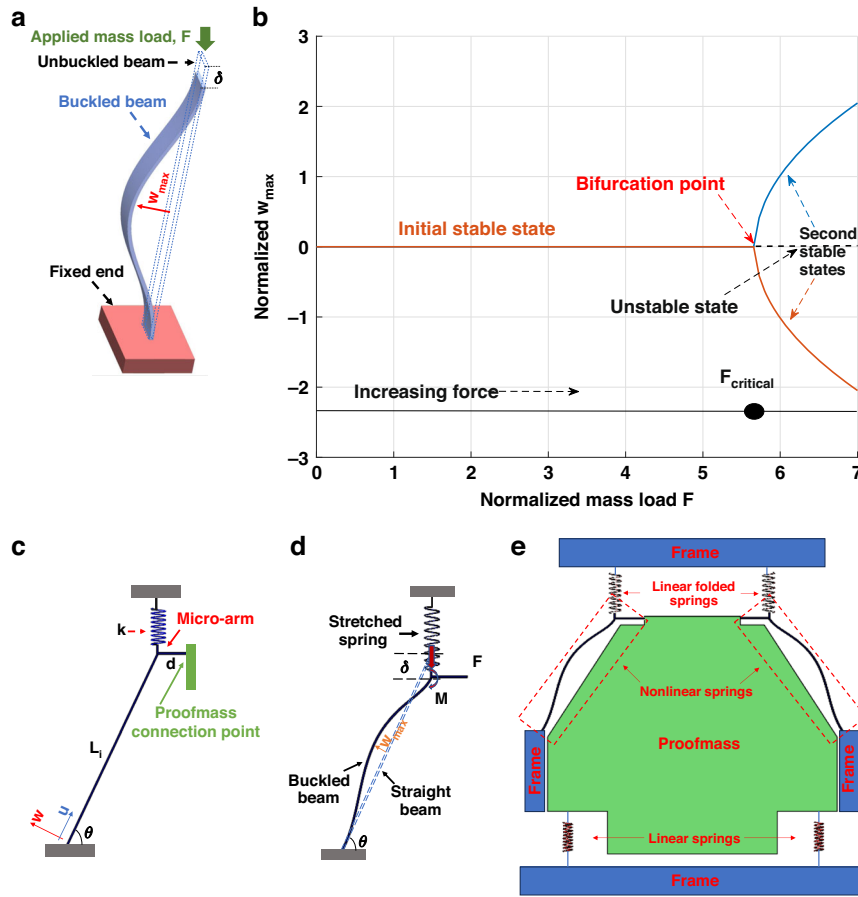


Fig. 1 The proposed nonlinear spring mechanism. **a** A buckled Euler beam with end shortening of δ in the y-direction, and displacement of $w_{\max}$ in the x-direction. **b** Static pitchfork-bifurcation curve. **c** The model of the eccentric buckling load and linear spring. **d** Free body diagram of the buckled beam and the stretched linear spring with applied axial and bending force and moment; and **e** A potential implementation of the nonlinear spring mechanism in a mass-spring system

buckled state. The mechanism is integrated into a mass-spring system, and is shown in Fig. 1e in its working state,

Comprehensive theoretical modeling and finite element analysis have been conducted to study and optimize the inclined beam's clamping angle and micro-arm length. Details of the foundational formulations are provided in the Supplementary Information (SI) document.

The governing equation of motion can be derived using Hamilton's principle:

$$\int_0^t (\delta T - \delta U + \delta W) dt = 0 \quad (1)$$

where T , U , and W represent the kinetic and potential energies, and the work done by external forces, respectively. As we are only interested in static motion, the kinetic term in Hamilton's principle is ignored.

The total potential energy of the system is composed of two terms: strain energy of the Euler-Bernoulli-based beam (U_s) and that of the spring (U_k): $U = U_s + U_k$. The

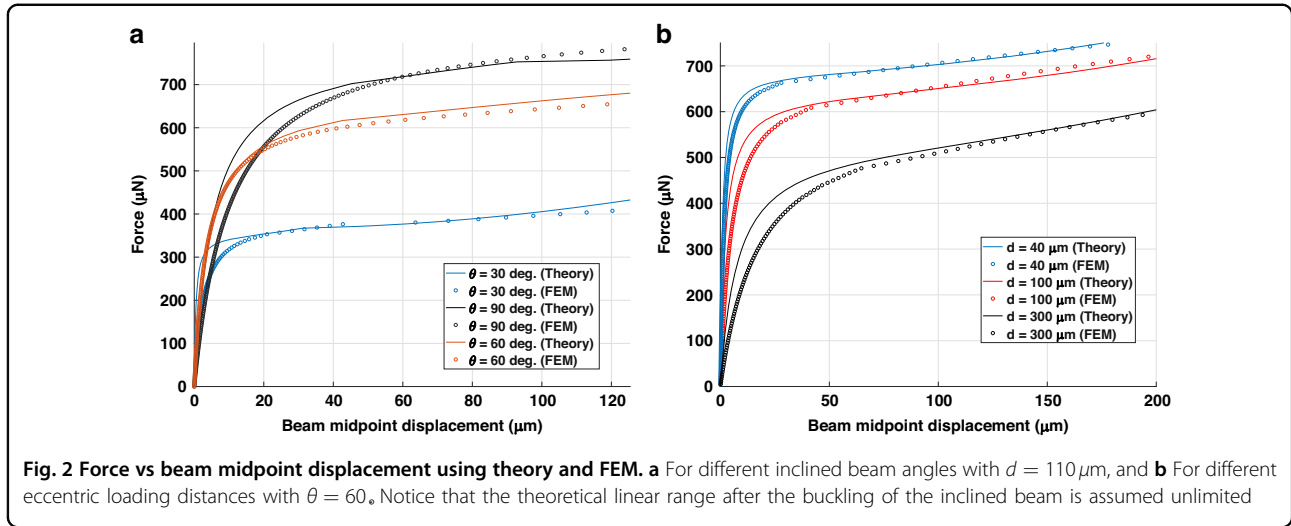
strain energy of an elastic beam can be obtained using (see SI document, Supplementary Eqs. (S2)–(6)):

$$U_s = \frac{1}{2} \int_0^{L_i} E_i I_i \left(\frac{\partial^2 w}{\partial x^2} \right)^2 dx + \frac{1}{2} \int_0^{L_i} E_i A_i \left(\frac{\partial u}{\partial x} + \left(\frac{\partial w}{\partial x} \right)^2 \right)^2 dx \quad (2)$$

where I and A represent the moment of inertia and cross-section area of the inclined beam, respectively. u and w are the axial and lateral deflections of the beam, respectively. The $\left(\frac{\partial w}{\partial x} \right)^2$ term in the strain energy is due to the Von-Karman (stretching nonlinearity)³⁶. In addition, the potential energy of the spring is obtained as follows:

$$U_k = \frac{1}{2} k \Delta x^2 = \frac{1}{2} k (u(L_i)^2 + w(L_i)^2) \quad (3)$$

The work done by the external force also is composed of two terms: one force is along the inclined beam axis, and



the other one is perpendicular to the inclined beam. The work done by the axial force in a transverse direction is³⁷:

$$W_{axial} = \frac{1}{2} \int_0^{L_i} F \sin \theta \cdot \left(\frac{dw}{dx} \right)^2 dx \quad (4)$$

While the work done by the force perpendicular to the beam is as follows:

$$W_{perp.} = F \cos \theta \cdot w(L_i) \quad (5)$$

The work done by moment $M = -F \cdot d$ is as follows:

$$W_M = \int_0^{L_i} M \cdot \frac{dw}{dx} dx \quad (6)$$

The nonlinear static equation of motion for the inclined beam can be obtained in terms of transverse displacement by substitution Eqs. (2–6) into Eq. (1), taking variations of each term, and applying integration by parts (details provided through Notes 1 and 2 in the SI document) as follows:

$$E_i I_i \frac{d^4 w}{dx^4} + (F \sin \theta + Fd) \frac{d^2 w}{dx^2} - \left[\frac{k E_i A_i}{2(E_i A_i + k L_i)} \int_0^{L_i} \left(\frac{dw}{dx} \right)^2 dx \right] \frac{d^2 w}{dx^2} = F \cos \theta \cdot \delta(x - L_i) \quad (7)$$

where $\delta(x - L_i)$ is the Dirac's delta function, indicating the external force is applied on the tip of the beam. The details of solving Eq. (7) using the Galerkin method are provided in the SI document.

Equation (7) can be solved numerically using MATLAB to obtain the inclined beam mid-point deflections under different conditions. Numerical modeling based on the Finite Element Analysis (FEA) in ANSYS was employed to

simulate system behavior and verify the modeling results, and to optimize the inclined beam angle (θ) and micro-arm length (d) under a preloading. Figure 2 compares the modeling and simulation results, demonstrating a good agreement between them. As shown, the force-displacement curve for each beam angle and micro-arm length exhibits two distinct regions. The initial portion corresponds to a low-sensitivity regime, where the structure behaves as a stiff spring with minimal displacement. The mild nonlinearity observed in this region arises from the eccentric loading, which introduces an initial deflection even before the critical buckling threshold is reached. Once the applied force exceeds the critical value, the system transitions into a high-sensitivity regime, where the effective stiffness drops significantly. The force-displacement curve for each beam angle is primarily influenced by two parameters. First, as evident from Eq. (7), the buckling threshold force is proportional to $\sin \theta$; therefore, adjusting the angle θ directly affects the axial force required to initiate buckling. Second, since the structure is intended to be fabricated on anisotropic silicon, varying the beam angle relative to the crystal orientation also changes the Young's modulus. These two factors together alter the required bias force and influence the overall shape of the force-displacement curve. This behavior is depicted in Fig. 2a for the fixed micro-arm length of $110 \mu\text{m}$. In addition, the micro-arm length contributes to the axial force component by introducing an additional moment, which facilitates the onset of buckling. As a result, increasing the length of the micro-arm reduces the required bias force needed to trigger buckling and enter the high-sensitivity region. This effect is studied in Fig. 2b for the fixed beam angle. This suggests that to reduce the bias force and displacement required for the softened stiffness, smaller d and θ values are preferred. However, decreasing the d value causes the

beam to behave closer to a standard buckling response (Fig. 1a, b), which quickly transitions to the post-buckling state. This would sharpen the slope of the curve in the post-buckling and may lead to damage or even failure of the mechanism. Also, choosing a smaller angle would require larger chip areas, which is often undesired. As a compromise between different factors, an inclination angle of $\theta = 60^\circ$ and $d = 300 \mu\text{m}$ were selected to fabricate a prototype.

Implementation of the proposed nonlinear spring in a MEMS sensor

Figure 3a shows the schematic of a micromachined spring accelerometer based on a pair of the proposed nonlinear spring mechanism. A proofmass is suspended by two sets of nonlinear springs on the top corners and two linear springs on the bottom corners. The system is designed to be symmetrical and move only in the vertical direction (sensing direction) by increasing torsional and rotational stiffnesses. The modal analysis of the design reveals the shapes and resonant frequencies of different modes, showing that the first resonant frequency corresponds to the sensing direction and is significantly lower than the others. This large separation ensures that the device is minimally responsive to vibrations or forces in other directions, thereby reducing cross-axis sensitivity (also see Supplementary Note 3 in SI document). In operation, inertial forces cause the beams to buckle, displacing the proofmass vertically until it enters the softening region. For gravimetry applications, this region is ideally centered around 1 g to ensure peak sensitivity near the device's operating bias. The proofmass displacement in the sensing direction (Δy) can have two parts: the first component occurs before buckling, where the micro-arm behaves as a stiff cantilever, resulting in an initial displacement of $\delta_0 = \frac{Mgd^3}{3EI}$, where Mg represents the weight of the proofmass. The second component arises from the buckling of the inclined beam, which causes a tip shortening of δ_b , determined with respect to the maximum displacement ($w_{\max}$) of the inclined beam as follows³⁸:

$$\frac{\delta_b}{L_i} \approx \frac{\pi^2}{4} \left(\frac{w_{\max}}{L_i} \right)^2 + \frac{3\pi^4}{64} \left(\frac{w_{\max}}{L_i} \right)^4 \quad (8)$$

Once the inclined beams buckle, the proofmass displacement will depend on the linear springs (δ_L). Therefore, the total displacement of the proofmass is: $\Delta y = \delta_0 + \delta_b + \delta_L$. It is noteworthy that if the d value is small, the initial displacement (δ_0) can be neglected, and the total displacement of the proofmass will primarily correspond to the shortening of the inclined beam caused by buckling and stiffnesses of the linear springs. Supplementary Table 1 presents the dimensions of the

nonlinear mechanism and linear springs used to design a MEMS prototype and study its force versus proofmass displacement curves and response to an input acceleration. Figure 3b illustrates the effect of varying beam angles on the force-displacement of the MEMS structure for a fixed micro-arm length of $d = 110 \mu\text{m}$. As can be seen, the design initially exhibits a high stiffness of $k = 730 \text{ N/m}$ for all beam angles. However, once buckling occurs, the mechanical effective stiffness of the system drops to only $k = 1.8 \text{ N/m}$, $k = 2.6 \text{ N/m}$, and $k = 2.2 \text{ N/m}$ for $\theta = 30^\circ$, 60° , and 90° , respectively. Additionally, the required bias force to push the MEMS structure into a high sensitivity working range is $F = 350 \mu\text{N}$, $600 \mu\text{N}$, and $720 \mu\text{N}$ for $\theta = 30^\circ$, 60° , and 90° , respectively, corresponding to an initial proofmass displacement of $\sim 3 \mu\text{m}$. Supplementary video SV1 shows the simulated response of the system to increasing input load until the inclined beams buckle.

The effect of the eccentric loading due to the micro-arm for a fixed beam angle of $\theta = 60^\circ$ is investigated in Fig. 3c. It is observed that as the micro-arm length value decreases, the bias force to push the MEMS sensor into the desired working range increases, while the initial proofmass displacement decreases significantly from $20 \mu\text{m}$ for $d = 300 \mu\text{m}$ to $\sim 3 \mu\text{m}$ for $d = 40 \mu\text{m}$. Reducing the micro-arm length d significantly lowers the system's mechanical stiffness. This is because smaller d values make the structure behave more like an ideal buckling beam, causing a rapid transition into the post-buckling regime. For example, the effective stiffness drops from 443 N/m to just 1.4 N/m when $d = 40 \mu\text{m}$. Although a shorter arm reduces the initial displacement of the proofmass, the abrupt transition is an undesired effect. Additionally, the upper limit of the post-buckling linear behavior is constrained by factors such as the beam's yield strength and possible contact between device components.

Although the proposed MEMS structure is intended for static applications, resonant frequency analysis remains essential, as both sensitivity ($S_{a_{in}}^y = \omega_0^{-2}$, with $\omega_0 = 2\pi f_0$ being the resonant frequency of the device) and noise floor (Eq. (12)) are closely tied to it. Lowering the resonant frequency without increasing the proofmass size is a key design goal for many micromachined inertial sensors. To assess the effectiveness of the nonlinear mechanism, the resonant frequency and corresponding displacement under varying input forces are analyzed in Fig. 3d, e. Initially, the structure exhibits high stiffness and resonant frequency, but once the input force exceeds the buckling threshold, the frequency drops significantly, enabling high-sensitivity behavior as shown in Fig. 3d. After buckling, the inclined beams contribute minimally to system stiffness, and the response is dominated by the linear springs. Figure 3e examines a case with the same inclined beam configuration but a higher aspect ratio for

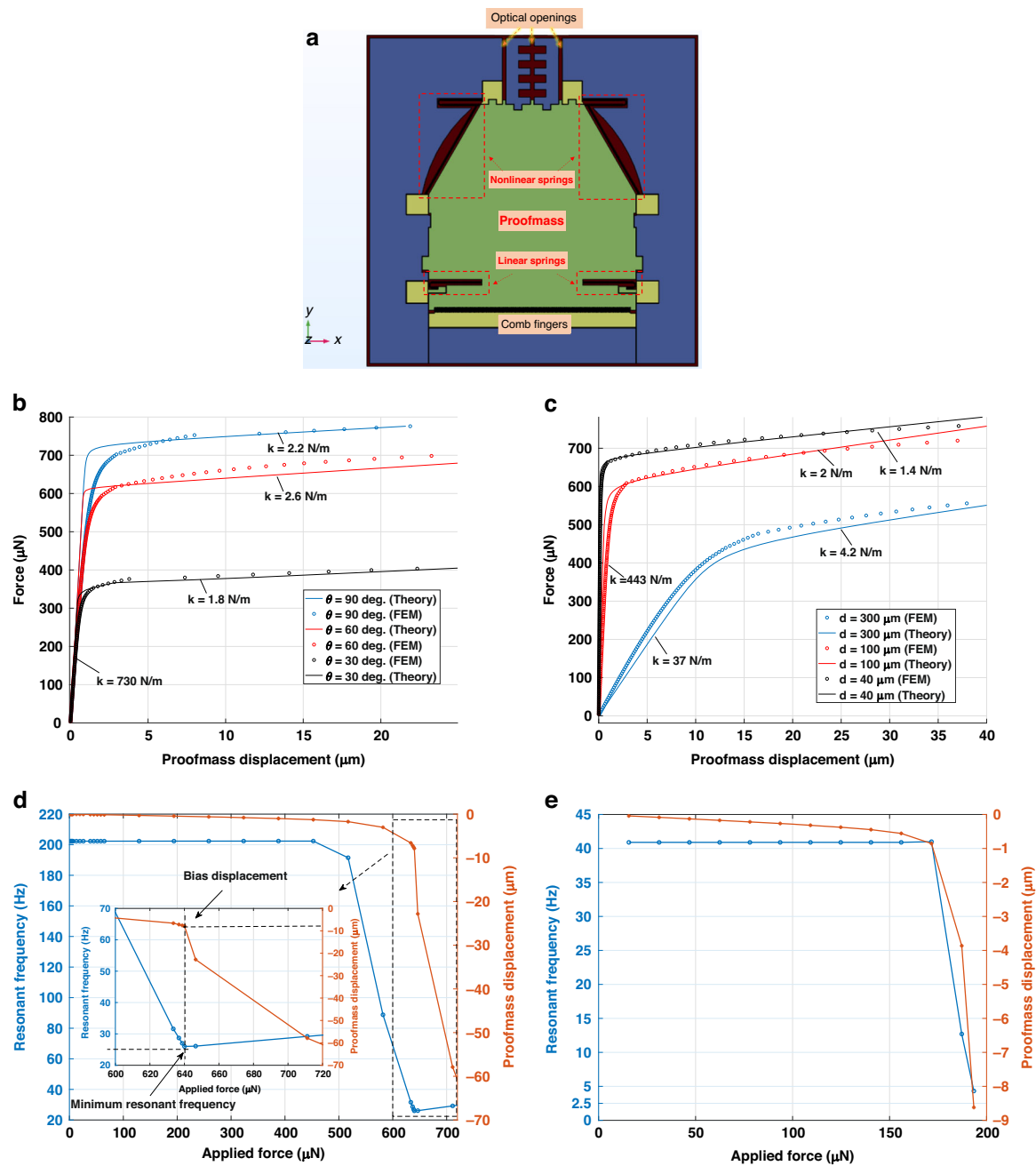
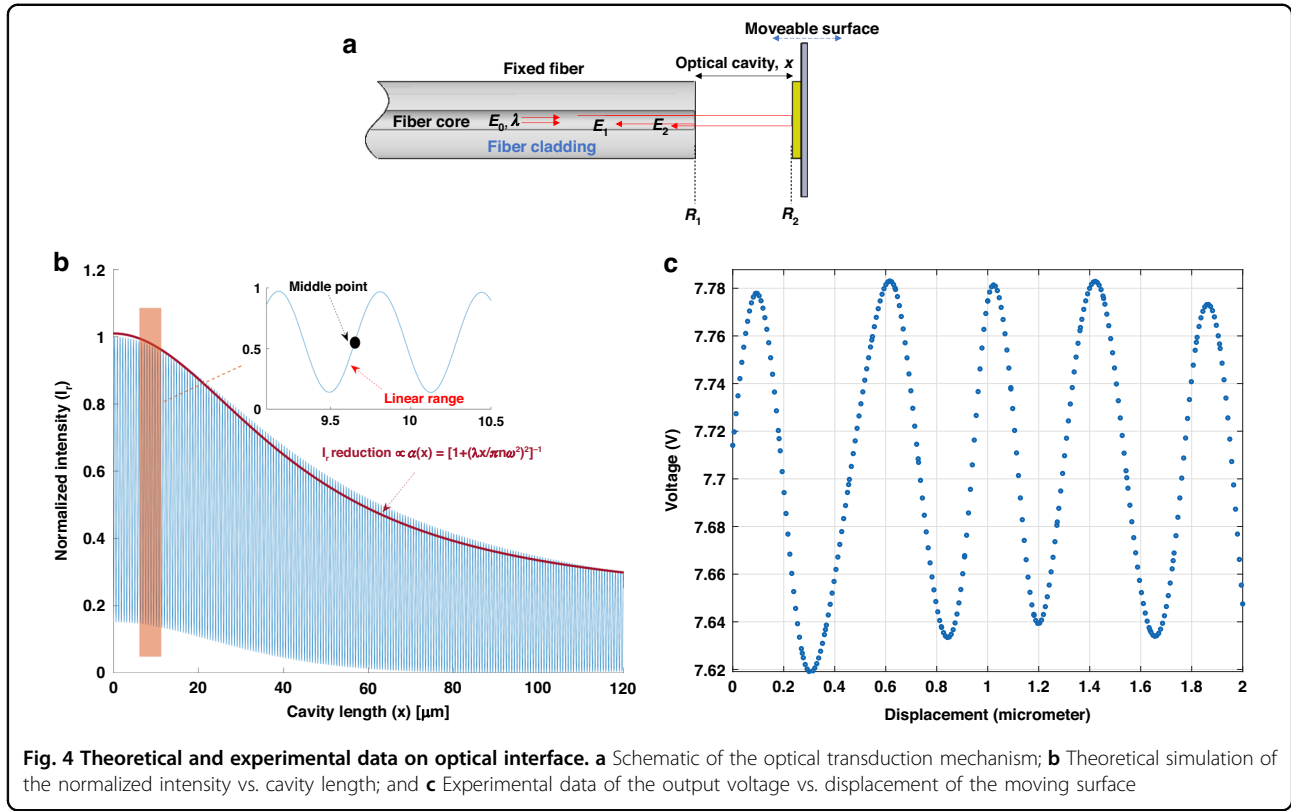


Fig. 3 Comparing model and simulations. **a** Schematic of the integrated proposed nonlinear spring mechanism into a MEMS structure with optical fiber opening, comb fingers, and nonlinear spring mechanism; **b** For different buckling beam angles with $d = 110 \mu\text{m}$; **c** For different eccentric loading distances with $\theta = 60^\circ$; **d** Applied force vs resonant frequency and bias displacement for $d = 110 \mu\text{m}$ and $\theta = 60^\circ$; and **e** Same as (d) with low-aspect-ratio linear springs

the linear springs, resulting in a resonant frequency of 4.3 Hz with only 9 μm of bias displacement. These results confirm that the bias displacement is primarily governed by the micro-arm length, the bias force by the beam angle, and the resonant frequency by the post-buckling stiffness of the linear springs. In all cases, the resonant frequency remains stable after buckling due to the constant spring

stiffness, extending the working range. Supplementary Video SV2 shows the simulated mode shapes and frequencies, which confirm that the dominant motion remains aligned with the sensing axis. Furthermore, the low resonant frequency and extended linear behavior suggest that the device could be used for measuring low-frequency dynamic signals, such as seismic activity.



Device interface

An on-chip optical interferometer is developed to measure the displacement of the fabricated MEMS proof mass in response to external accelerations. Figure 4a illustrates the mechanism of the interferometer, where an optical cavity is formed between the proofmass side as the reflector and a fiber's end-face affixed to the chip. As the applied inertial force changes, the cavity length between the fiber end-face and the reflector changes, hence shifting the phase-difference between the transmitted and the reflected light. Amplitudes of the light waves reflected from the reflector surface and the fiber end-face can be expressed as³⁹:

$$\begin{aligned} E_1 &= \sqrt{R_1}E_0 \\ E_2 &= -(1 - R_1)\sqrt{R_2}\sqrt{\alpha(x)}E_0 \\ E_3 &= -(1 - R_1)R_2\sqrt{R_1}\sqrt{\alpha(2x)}E_0 \end{aligned} \quad (9)$$

where E_0 , E_1 , E_2 , and E_3 are the incident light amplitude, the amplitude of the light reflected from the fiber end-face, the amplitude of the light reflected from the reflector, and the amplitude of the light reflected twice from the reflector, respectively. $\alpha(x)$ is the light intensity attenuation coefficient in the optical cavity. In addition, R_1 and R_2 are the reflection index of the fiber end-face and the reflector, respectively. By considering the phase shifts of the interfering light, the total reflected light quantity

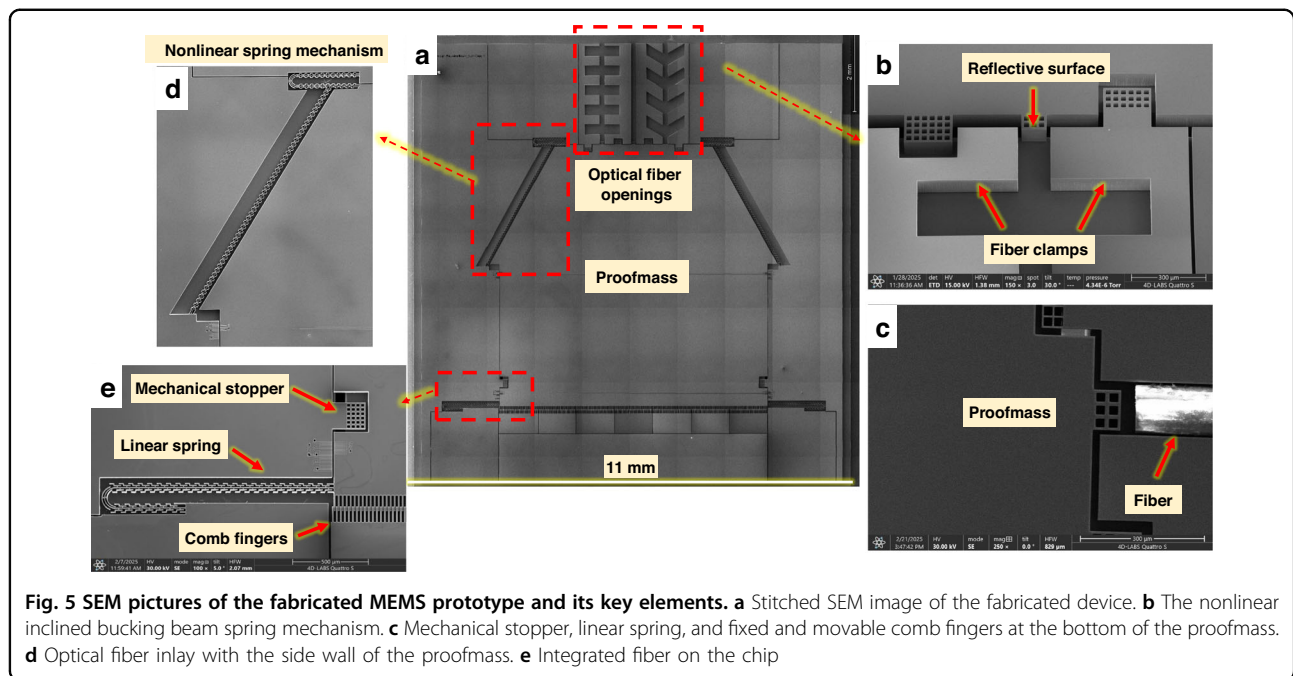
can be expressed as:

$$E_r = E_1 + E_2e^{i\delta} + E_3e^{i2\delta} \quad (10)$$

where $\delta = 2\pi nx/\lambda$ is the phase shift produced between the light reflected from the fiber end-face and the reflector surface, n is the reflection index of the air, x is the optical cavity length, and λ is the light source wavelength. The light intensity can be obtained by using the formula of $I_r(d, \lambda, n) = E_r E_r^*$, where E_r^* is the conjugate of the E_r . This process can be simplified by considering that the fiber end-face reflection index is small ($R_1 = 0.04$). Hence, the light amplitudes reflected from the fiber end-face can be neglected compared to the incident light amplitude and the light amplitude reflected from the reflector. Consequently, the optical relative intensity can be expressed as:

$$\begin{aligned} I_r &= R_1E_0^2 + (1 - R_1)^2R_2\alpha(x)E_0^2 - 2(1 - R_1) \\ &\quad \sqrt{R_1R_2}\sqrt{\alpha(x)}E_0^2 \cos \delta \end{aligned} \quad (11)$$

As the above equation indicates, the total reflected intensity varies with the optical cavity length, x , the wavelength, λ , and the reflective index of the air, n . Figure 4b shows the numerical solution to Eq. (11) in MATLAB to investigate the effect of the cavity length on the total reflected intensity, indicating that a smaller cavity length results in a higher



intensity. The experimental results, shown in Fig. 4c, indicate that the output voltage varies periodically with displacement.

Experimental results

Figure 5 shows the scanning electron microscope (SEM) images of the fabricated MEMS prototype, with an external chip area of $11 \times 11 \text{ mm}^2$. The left side of the image highlights the nonlinear and linear springs, while the right side displays the optical interferometer features. An SEM image of the proposed nonlinear spring mechanism is shown in Fig. 5b. Also, as shown in Fig. 5c, several mechanical stoppers are placed around the proofmass to prevent structural failure due to external shocks and to limit the movement of the proofmass in undesired directions. Additionally, a set of comb fingers with a length of $100 \mu\text{m}$, gap of $5 \mu\text{m}$, and an initial overlap of $15 \mu\text{m}$, although not utilized in this work, are positioned at the bottom of the proofmass for actuation purposes. These comb fingers are designed to push the proofmass into the optical output's linear range and facilitate future closed-loop operation. Figure 5d shows the zoomed-in view of the optical fiber opening with a cross-sectional view of the proofmass wall. To control the initial distance between the fiber and the proofmass, three different optical openings were designed. Among them, the central opening does not include a fiber stopper, allowing for a minimized initial optical cavity length to enhance optical sensitivity. The integrated optical fiber on the chip is shown in Fig. 5e. Due to the fabrication and constraints on fiber integration on the chip, an initial cavity length of $34 \mu\text{m}$ was achieved and utilized.

Figure 6a shows the schematic of the complete optical interferometer setup to characterize the fabricated MEMS sensor. The optical setup consists of a laser source (Toptica DL Pro 850)⁴⁰ with a fixed central wavelength of $\lambda = 850 \text{ nm}$ and a maximum output power of 65 mW whose output is connected to a circulator (Thorlabs, WMC3L1F)⁴¹. The output of the circulator is coupled to the wafer that is affixed to the MEMS chip and the reflected light from the circulator is measured with a photodetector (Thorlabs, FDS02)⁴², which is connected to a photodiode amplifier (Thorlabs, PDA200C)⁴³. The amplifier is connected to a data acquisition device (USB 5801)⁴⁴ to collect the signal data and transfer it to a computer for monitoring the proofmass displacement. To fix the fiber to the chip, its cladding is first stripped and it is passed through a ferrule to allow for the proper alignment with the proofmass. Once proper alignment is achieved, the fiber is glued to the chip. The experimental setup is shown in Fig. 6b, with a view of the customized optical package containing the MEMS sensor, mounted on a high-precision tilt stage as shown in Fig. 6c.

The performance of the spring accelerometer was evaluated using a tilting method by mounting the MEMS device on a high-precision tilt stage (Aerotech ARMS 200)⁴⁵ with a resolution of $10 \mu\text{-degrees}$ as shown in Fig. 6c. Initially, the MEMS device was mounted on a tilt stage such that its sensitive axis was perpendicular to the stage surface. It was then gradually tilted while monitoring the output response to confirm the presence of nonlinearity in the system, as shown in the first half of Fig. 6e. The experiment was subsequently repeated by tilting the device back to its original position, as shown in the second half of Fig. 6e (see

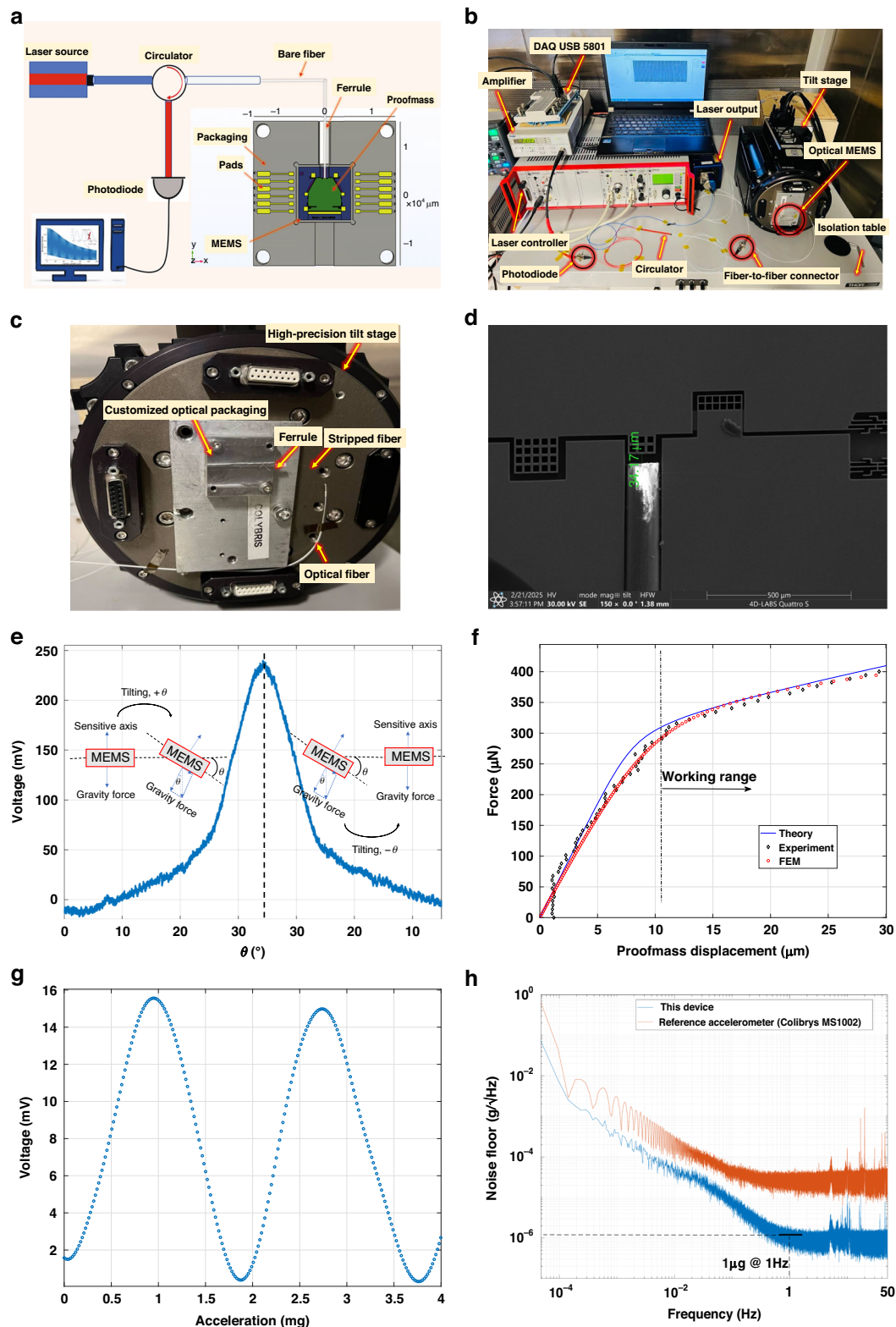


Fig. 6 Performance evaluation of the proposed nonlinear spring mechanism integrated into an optical MEMS prototype. **a** Schematic of the optical displacement interferometer system. **b** Experimental setup. **c** Customized optical package mounted on the high precision tilt stage. **d** An SEM image of the aligned fiber with proofmass cross-section with an initial optical cavity of $34\ \mu\text{m}$. **e** Measured proofmass displacement as the device was tilted from $\sim 55^\circ$ to 90° ($\Delta\theta \approx 35^\circ$) and back, showing the proofmass's drop under preset load and restoration afterwards. **f** Comparison of force vs proofmass displacement from analytic and numerical models against values obtained experimentally. **g** Measured output voltage from the optical interferometer; and **h** Acceleration noise floor of the MEMS compared to a high-performance accelerometer in the lab without environment control

Table 1 Performance comparison of the implemented nonlinear spring mechanism in a MEMS prototype with others

Ref.	Readout	Spring type	Bias displ. (μm)	Chip size ($L \times W$, mm^2)	Bias force (mN)	Noise floor ($\frac{\text{ng}}{\sqrt{\text{Hz}}}$)	Quality factor	Sensitivity improvement	Dynamic Range (mg)	Resonance frequency (Hz)
[16]	Optical shadow	Curved beams	800	20×20	0.42	40	80	55×	1	2.3
[21]	Optical shadow	Curved beams	500	25×25	3.3	8	1200	940×	8	3.1
[19]	Capacitive	Curved beams	600	15×15	0.37	17 ^a	12	18×	40	8.7
[23]	Optical shadow	Curved beams	350	20×20	0.7	9.2	/	10.5×	24	8.5
[25]	Optical grating	Free-form beams	1000	35×35	2.2	1.1	/	/	10	1.7
[24]	Capacitive	Curved beams	1.6	4.2×4.9	0.018	21,300	4	1.11×	/	2.3 kHz
This work	Optical interferometer	Inclined beams	10	11×11	0.3	1000/45 ^b	1	78×	167	21 ^c

^aTheoretical calculation for $Q \approx 12$ ^bMeasured/Theoretical calculation for $Q \approx 1$ ^cSimulated

Supplementary Video SV3, which shows the tilt response of the device on the chip as observed under a microscope). Figure 6f compares the proofmass displacement derived from the theoretical model, experimental data, and FEM simulations in ANSYS. The experimental values were calculated from the recorded output voltage from the interferometer in conjunction with the calibrated optical intensity factor. As can be seen, there is good agreement among these results. The sensitivity of the MEMS device to an input acceleration was assessed by gradually tilting the device. The fabricated MEMS prototype demonstrates a wide range of linear behavior after buckling, indicating an extended dynamic range for the proposed nonlinear mechanism. Figure 6g shows the output voltage measured from the photodetector, which reflects the periodic interference signal generated by the on-chip optical cavity as the input load varies, demonstrating a sensitivity of $194^\circ/\text{mg}$. Finally, the noise floor of the proposed MEMS was measured in a lab environment with significant background noise and without temperature or pressure control. The tilt method was used to place the proofmass in the optical linear response range for maximum optical sensitivity. Under these conditions, a minimum noise level of $1 \mu\text{g}/\sqrt{\text{Hz}}$ was measured as shown in Fig. 6h. However, based on our earlier work in our laboratory, we believe this limit is due to the environmental vibration noise⁴⁶. The thermal noise equivalent acceleration of the proposed MEMS prototype can be calculated using:

$$NEA_{\text{mechanical}} = \sqrt{4k_B T \frac{\omega_0}{mQ}} \quad (12)$$

where k_B and T are the Boltzmann constant and measurement temperature, respectively, $m = 60 \text{ mg}$ is the device weight, and $f_0 = \omega_0/2\pi = 21 \text{ Hz}$ is the simulated fundamental frequency of the proposed MEMS device. With an estimated quality factor of $Q \approx 1$, the

equivalent acceleration noise of the devices is estimated to be better than $45 \text{ ng}/\sqrt{\text{Hz}}$.

Discussion

The experimental results validate the effectiveness of the proposed nonlinear spring mechanism in achieving stable stiffness softening with low bias force and displacement. Table 1 summarizes the performance comparison of the proposed nonlinear spring mechanism integrated into an optical MEMS prototype and existing nonlinear MEMS accelerometers. Most MEMS sensors listed in Table 1 utilize pre-curved beams to enhance mechanical sensitivity, which typically requires large bias forces and displacements. Our proposed mechanism requires a small bias force of 0.3 mN to reach the stiffness softening region, representing an $11\times$ reduction compared to the 3.3 mN reported in ref. 21. The spring mechanism also reduces the bias displacement to $10 \mu\text{m}$, representing improvements of $35\times$, $50\times$, $60\times$, and $80\times$ compared to the bias displacements of $\sim 350 \mu\text{m}$ ²³, $\sim 500 \mu\text{m}$ ²¹, $\sim 600 \mu\text{m}$ ¹⁹, and $\sim 800 \mu\text{m}$ ¹⁶, respectively. At such low bias displacements, capacitive sensing becomes a viable readout mechanism for the proposed structure. However, achieving the required sensitivity would require a large number of comb fingers, which introduces both design and fabrication complexity. Additionally, capacitive sensors are vulnerable to particulate contamination; any particle lodged between comb fingers could shorten the structure or render the device inoperable. As a proof of principle, using an optical interface allowed us to maintain focus on the mechanical performance of the device. A limitation of using curved beams is the sudden transition between bistable positions, which results in a limited dynamic range. Our proposed nonlinear mechanism shows extended linear behavior after buckling, which results in a large dynamic range of 167 mg (limited by the buckled beams touching the frame of the device).

These improvements enable a die area reduction by an order of magnitude, resulting in a compact $11 \times 11 \text{ mm}^2$ form factor compared to other works, such as²¹ and²³ which report die dimensions of $25 \times 25 \text{ mm}^2$ and $20 \times 20 \text{ mm}^2$, respectively. This smaller size enables scalable mass production, simplifies die handling during microfabrication and testing, enhances application flexibility, and facilitates easier integration into various systems. Supplementary Note 4 shows pictures of the proposed prototype microfabrication at the wafer level. Although the resonant frequency of our fabricated MEMS prototype is higher than the designs reported in Table 1, as discussed in Fig. 3e, our design can achieve a resonant frequency of 4.3 Hz by selecting longer linear springs while maintaining a small bias displacement. Also, the utilized high-sensitivity optical readout interferometer compensates for the low mechanical sensitivity of the MEMS sensor. In addition, our proposed spring mechanism demonstrates a sensitivity improvement compared to its initial sensitivity by a factor of $78\times$, which is higher than the improvements reported in refs. ^{16,19,23}, and²⁴ but lower

than the $940\times$ improvement reported in ref. ²¹. However, our measured noise floor in a standard laboratory environment without temperature and pressure control is $1 \mu\text{g}/\sqrt{\text{Hz}}$ limited by the environmental noise vibrations, as compared to the theoretical value of $45 \text{ ng}/\sqrt{\text{Hz}}$ based on a conservative quality factor estimate of $Q \approx 1$. This work demonstrates a promising approach for advancing MEMS sensor design, particularly in applications requiring high sensitivity, large dynamic range, and miniaturization. Future work will focus on integrating this nonlinear mechanism into a capacitive-based feedback MEMS sensor with an on-chip optical readout and temperature monitoring, further exploring its potential for performance enhancement and real-world applications.

Materials and methods

Fabrication process flow

The proposed optical MEMS prototype is fabricated in the SFU 4D LABS cleanroom. The fabrication process flow, which is based on a Silicon-On-Insulator (SOI) wafer, is shown in Fig. 7a–f. The external dimension of the chip is

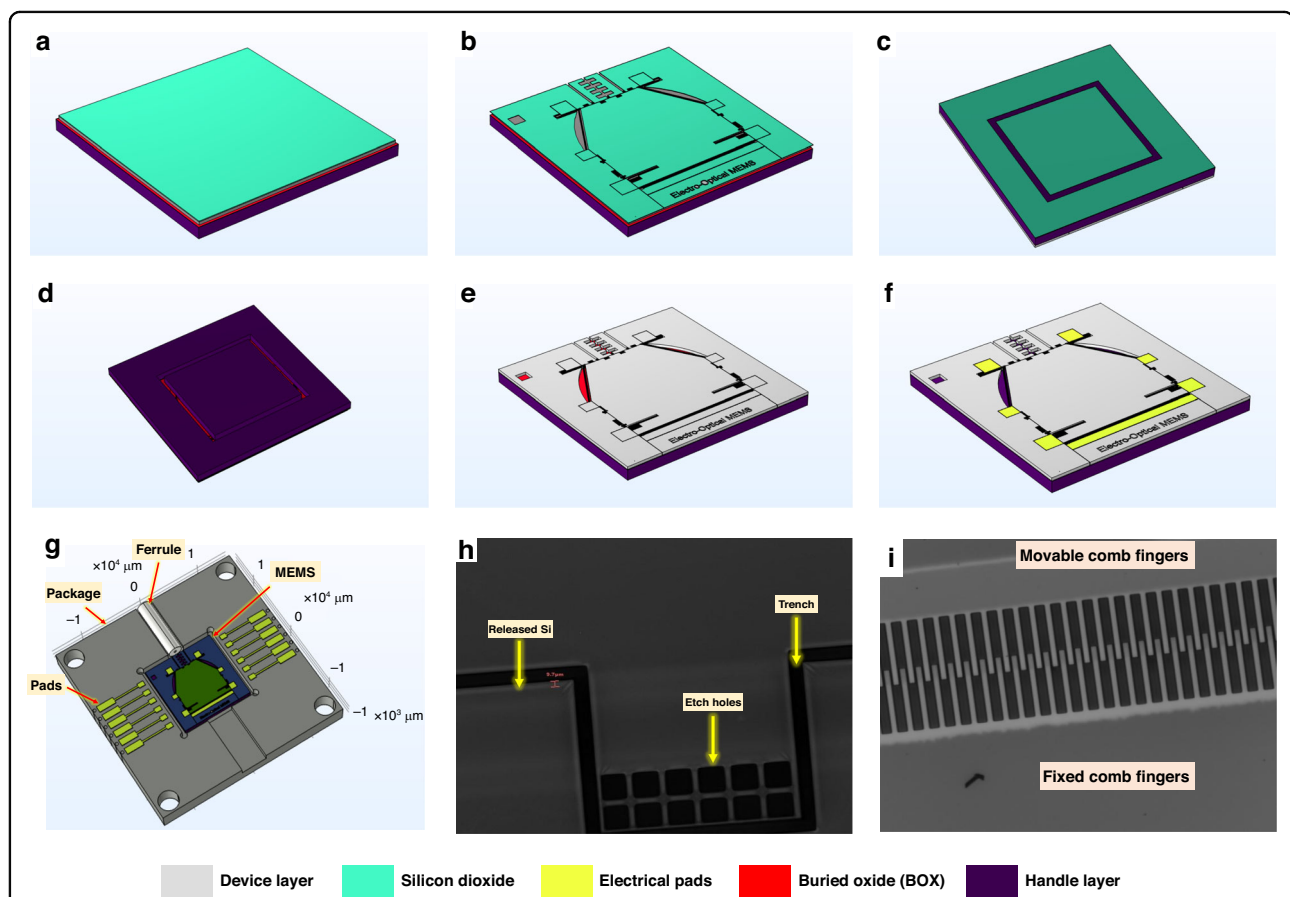


Fig. 7 Fabrication process flow of the MEMS prototype. **a** SOI wafer, **b** Trenches patterned using RIE on the silicon oxide, **c** Same as (b) on the backside, **d** DRIE of the handle layer, **e** DRIE of the device layer, **f** Vapor HF wet etching for BOX layer release then metal layer deposition for wire bonding, **g** Customized optical package, **h** trenches and etch holes, and **i** comb fingers in a released device

11 mm $\times$ 11 mm. The fabricated chip comprises two layers: a device layer with a thickness of 100 μm that is used to etch the proofmass, springs, and optical and electrical features, and a handle layer with a thickness of 575 μm which is partially kept and used for an additional mass. The fabrication begins with the proper cleaning of the SOI wafer. Then, a 2 μm layer of silicon dioxide is thermally grown on top of the device layer (a). Through a photolithography process, the oxide layer is patterned to be used as the hard mask for the subsequent silicon etching. The exposed area of the oxide layer is etched using Reactive-Ion-Etching (RIE) (b). The same process is used to etch the backside oxide layer (c) followed by etching the handle wafer through a Deep-Reactive-Ion-Etching (DRIE) step (d). Then, the patterned frontside oxide layer is used as a masking layer to etch the device silicon layer in another DRIE step (e). Finally, the device is released via vapor hydrofluoric acid (VHF) to release the moving parts from the buried oxide (BOX) layer (f). Once the fabrication is completed, the MEMS prototype is mounted in customized packaging for performance

testing, as shown in Fig. 7g. Figure 7h, i shows infrared images of the released device parts, including the trenches, etch holes, and comb fingers, where the light parts in these pictures show the released part of the silicon from silicon dioxide after VHF etching.

Optical intensity calibration

Figure 8a shows the schematic of the optical intensity calibration setup. In this experiment, as shown in Fig. 8b, a step-shaped sample with the same material as the proofmass was fabricated using DRIE with a step height of $\sim 100 \mu\text{m}$ to calibrate the optical intensity reflection amplitude. As shown in Fig. 8a, the sample is bonded to a sample carrier, which is attached to a micro-positioner. The micro-positioner is located under a vertical stand, where a fiber is properly aligned to the sample using a fiber holder. The distance between the fiber end face and the sample surface, acting as a reflector, can be adjusted. Initially, the fiber faces a flat surface of the sample, establishing the reference distance. By adjusting the

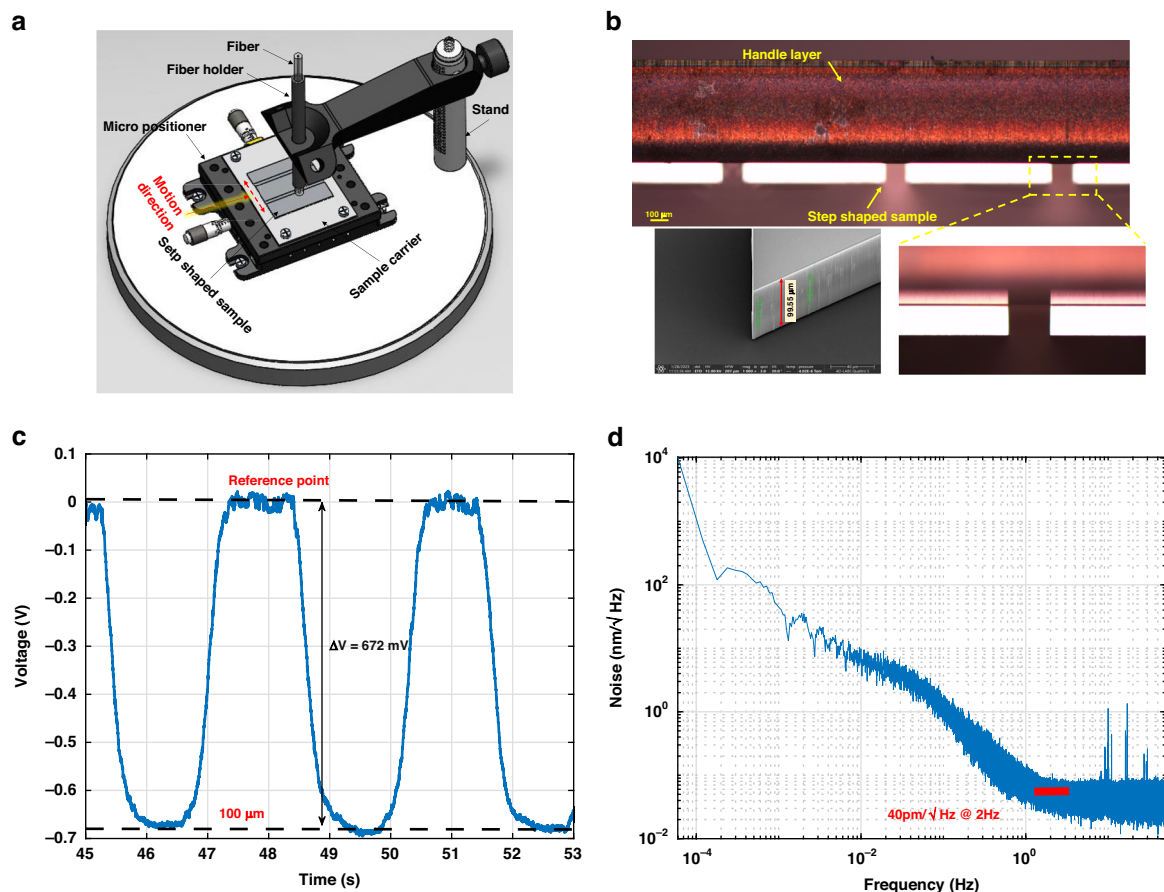


Fig. 8 Experiments of optical intensity calibration. **a** Schematic of the used setup, **b** A microscopic cross-sectional view of the fabricated step-shaped sample with an SEM image of the cross-section, **c** Output voltage for the height change, and **d** Displacement noise floor of the optical readout sensor

micro-positioner, the distance between the fiber and the sample changes when the fiber faces the step-shaped region. The output voltage corresponding to different positions of the sample is shown in Fig. 8c, where 0 V indicates the fiber facing the flat surface of the sample, and 672 mV corresponds to the fiber being within the step-shaped region. This results in an intensity calibration factor of $102 \mu\text{m}/672 \text{ mV}$. This factor is used to obtain the proofmass displacement to an external acceleration. Then, the silicon sample was positioned in a stationary setup with a fixed fiber facing it to measure the displacement noise floor of the optical sensor. Figure 8d presents experimental results, showing a displacement noise floor of $40 \text{ pm}/\sqrt{\text{Hz}}@2 \text{ Hz}$, limited by environmental noise.

Acknowledgements

This research was financially supported by the Canadian Space Agency (CSA). We would like to acknowledge CMC Microsystems, the manager of the FABrIC project, which is funded by the Government of Canada, for providing CAD tools and test equipment that facilitated this research. This work made use of the 4D LABS core facility at Simon Fraser University, supported by the Canada Foundation for Innovation (CFI), British Columbia Knowledge Development Fund (BCKDF), and Pacific Economic Development Canada (PacifiCan).

Author contributions

P.F. performed the theoretical modeling, simulations, layout designs, fabrication, packaging, conducting the experiments and data analysis, writing the original draft, and reviewing and editing drafts. M.S.H. contributed to the fabrication, technical discussion on the layout design, experiments, packaging and data analysis. M.K. contributed to the fabrication and device layout. Ph.F. provided technical guidance and modeling assistance. B.B. provided supervision, technical guidance, validation, funding, and assisted with editing the manuscript. All authors reviewed and commented on the manuscript.

Conflict of interest

The authors declare no competing interest.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41378-025-01066-3>.

Received: 2 April 2025 Revised: 16 June 2025 Accepted: 12 August 2025
Published online: 24 October 2025

References

- Ullah, P., Ragot, V., Zwahlen, P. & Rudolf, F. A new high performance sigma-delta MEMS accelerometer for inertial navigation. In *2015 DGON Inertial Sensors and Systems Symposium (ISS)* 1–13 (IEEE, 2015).
- Zhu, L. et al. Development of a high-sensitivity wireless accelerometer for structural health monitoring. *Sensors* **18**, 262 (2018).
- Amini, B. V. & Ayazi, F. Micro-gravity capacitive silicon-on-insulator accelerometers. *J. Micromech. Microeng.* **15**, 2113–2120 (2005).
- Mustafazade, A. et al. A vibrating beam MEMS accelerometer for gravity and seismic measurements. *Sci. Rep.* **10**, 10415 (2020).
- Lognonné, P. et al. SEIS: insight's seismic experiment for internal structure of Mars. *Space Sci. Rev.* **215**, 12 (2019).
- Edalatfar, F. et al. A wideband, low-noise accelerometer for sonar wave detection. *IEEE Sens. J.* **18**, 508–516 (2018).
- Wu, W. et al. A precise spacing-control method in MEMS packaging for capacitive accelerometer applications. *J. Micromech. Microeng.* **28**, 125016 (2018).
- Pike, W. T., Standley, I. M., Calcutt, S. B. & Mukherjee, A. G. A broad-band silicon microseismometer with 0.25 NG/rtHz performance. In *2018 IEEE Micro Electro Mechanical Systems (MEMS)* 113–116 (IEEE, 2018).
- Qu, Z. et al. 2.4 ng/ $\sqrt{\text{Hz}}$ low-noise fiber-optic MEMS seismic accelerometer. *Opt. Lett.* **47**, 718 (2022).
- Firoozy, P., Haghighi, M. S., Kanygin, M., Esmaeili, E. & Bahreyni, B. A MEMS gravimeter with buckling-beam nonlinear springs for enhanced sensitivity and dynamic range. In *2024 IEEE SENSORS* 1–4 (IEEE, 2024).
- Fang, W. et al. Accurate mechanical-optical theoretical model of cross-axis sensitivity of an interferometric micro-optomechanical accelerometer. *Appl. Opt.* **61**, 3201 (2022).
- Hussein, H., Wang, C., Amendoeira Esteves, R., Kraft, M. & Fariborzi, H. Near-zero stiffness accelerometer with buckling of tunable electrothermal microbeams. *Microsyst. Nanoeng.* **10**, 43 (2024).
- Wang, C. et al. Design of a capacitive MEMS accelerometer with softened beams. *Micromachines* **13**, 459 (2022).
- Haghighi, M. S., Firoozy, P., Kanygin, M., Esmaeili, E. & Bahreyni, B. A new nonlinear micromechanical structure used in MEMS inertial sensors. In *2024 IEEE SENSORS* 1–4 (IEEE, 2024).
- Li, F. et al. Novel area-changed capacitive methods for simultaneous displacement transducing and force balance in a nano-g MEMS accelerometer. *J. Microelectromechanical Syst.* **33**, 12–20 (2024).
- Middlemiss, R. P. et al. Measurement of the Earth tides with a MEMS gravimeter. *Nature* **531**, 614–617 (2016).
- Prasad, A. et al. A 19 day earth tide measurement with a MEMS gravimeter. *Sci. Rep.* **12**, 13091 (2022).
- Boom, B. A. et al. Nano-G accelerometer using geometric anti-springs. In *2017 IEEE 30th International Conference on Micro Electro Mechanical Systems (MEMS)* 33–36 (IEEE, 2017).
- El Mansouri, B. et al. High-resolution MEMS inertial sensor combining large-displacement buckling behaviour with integrated capacitive readout. *Microsyst. Nanoeng.* **5**, 60 (2019).
- Zhang, H., Wei, X., Ding, Y., Jiang, Z. & Ren, J. A low noise capacitive MEMS accelerometer with anti-spring structure. *Sens. Actuators Phys.* **296**, 79–86 (2019).
- Tang, S. et al. A high-sensitivity MEMS gravimeter with a large dynamic range. *Microsyst. Nanoeng.* **5**, 45 (2019).
- Tang, S. et al. A MEMS gravimeter qualified for Earth tides measurement. In *2019 20th International Conference on Solid-State Sensors, Actuators and Microsystems & Eurosensors XXXIII (TRANSDUCERS & EUROSENSORS XXXIII)* 499–502 (IEEE, 2019).
- Weidong, W. et al. A low-noise MEMS gravimeter capable of attaining stable low resonant frequency at various tilt angles through adjusting curved beams' width. *IEEE Trans. Electron Devices* 1–9, <https://doi.org/10.1109/TED.2025.3532904> (2025).
- Xiong, R. et al. A miniaturized MEMS accelerometer with anti-spring mechanism for enhancing sensitivity. *Microsyst. Nanoeng.* **11**, 1–15 (2025).
- Wu, S. et al. A μGal MOEMS gravimeter designed with free-form anti-springs. *Nat. Commun.* **16**, 1786 (2025).
- Furubayashi, Y. et al. 10.2 A 22ng/ $\sqrt{\text{Hz}}$ 17mW MEMS accelerometer with digital noise-reduction techniques. In *2019 IEEE International Solid-State Circuits Conference - (ISSCC)* 182–184 (IEEE, 2019).
- Pike, W. T. et al. Design, fabrication and testing of a micromachined seismometer with NANO-G resolution. In *TRANSDUCERS 2009 - 2009 International Solid-State Sensors, Actuators and Microsystems Conference* 668–671 (IEEE, 2009).
- Ahmed, Z., Duruaku, C., Edalatfar, F., Moallem, M. & Bahreyni, B. A low-noise micromachined accelerometer with reconfigurable electrodes for resonance suppression. *Micromachines* **14**, 1188 (2023).
- Ouyang, H. et al. Nano-g accelerometer with differential Fabry-Pérot interferometer for low-frequency noise suppression. *Sens. Actuators Phys.* **346**, 113874 (2022).
- Pan, D., Liang, Y., Zhang, Z. & Wu, Z. Design and dynamics of a cantilevered bistable buckled piezoelectric beam for vibrational energy harvesting. *Mech. Syst. Signal Process.* **224**, 112013 (2025).
- Fu, H. et al. Morphable 3D mesostructures and microelectronic devices by multistable buckling mechanics. *Nat. Mater.* **17**, 268–276 (2018).
- Kanj, A., Ferrari, P., Van Der Zande, A. M., Vakakis, A. F. & Tawfik, S. Ultra-Tuning of nonlinear drumhead MEMS resonators by Electro-Thermoelastic buckling. *Mech. Syst. Signal Process.* **196**, 110331 (2023).
- Yao, A. & Hikiara, T. Reprogrammable logic-memory device of a mechanical resonator. *Int. J. Non-Linear Mech.* **94**, 406–416 (2017).

34. Najar, F., Ghommem, M., Kocer, S., Elhady, A. & Abdel-Rahman, E. M. Detection methods for multi-modal inertial gas sensors. *Sensors* **22**, 9688 (2022).
35. Park, S. & Hah, D. Pre-shaped buckled-beam actuators: Theory and experiments. *Sens. Actuators Phys.* **148**, 186–192 (2008).
36. Doyle, J. F. Introduction. In *Spectral Analysis of Nonlinear Elastic Shapes* (ed. Doyle, J. F.) 1–5 (Springer International Publishing, 2020).
37. Rao, S. S. *Vibration of Continuous Systems* (John Wiley & Sons, 2019).
38. Leandri, G. L., Sunderland, A., Winterflood, J. & Ju, L. Six degrees of freedom vibration isolation with Euler springs. *Rev. Sci. Instrum.* **92**, 025122 (2021).
39. Qu, Z. et al. A nano-g MOEMS accelerometer featuring electromagnetic force balance with 157-dB dynamic range. *IEEE Trans. Ind. Electron.* **71**, 6418–6426 (2024).
40. DL pro. *TOPTICA Photonics AG* <https://www.toptica.com/products/tunable-diode-lasers/ecdl-dfb-lasers/dl-pro> (2025).
41. Thorlabs - WMC3L1F Wideband Multimode Circulator. <https://www.thorlabs.com/thorproduct.cfm?partnumber=WMC3L1F> (2025).
42. Thorlabs - FDS02 Si Photodiode. <https://www.thorlabs.com/thorproduct.cfm?partnumber=FDS02> (2025).
43. Thorlabs - PDA200C Photodiode Amplifier. <https://www.thorlabs.com/thorproduct.cfm?partnumber=PDA200C> (2025).
44. USB-5801. https://www.advantech.com/en/products/1-2mlkno/usb-5801/mod_b22c7b89-f13c-4604-9b48-cef24cd927a2 (2025).
45. Aerotech ARMS200 Single Axis Rotary Motion Simulator. <https://www.aerotech.com/product/arms-mechanical-bearing-direct-drive-rotary-stage/> (2025).
46. Kanygin, M. et al. A linear capacitive gravimeter with pseudo-differential electrodes capable of measuring Earth tides. In *2025 IEEE 38th International Conference on Micro Electro Mechanical Systems (MEMS)* 912–915 (2025).